

Electrical Services Design

Engineering Technology

May, 2026

Electrical Services Design (A05930)

Short Title:	Electrical Services Design	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electrical Engineering	
Author	CLAIRE.FITZPATRICK	
Credits	5	Level: Intermediate

Description of Module / Aims

The aim of this module is to provide the knowledge and skills necessary to design and layout the electrical services and systems for a modern building/ plant.

Programmes

	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	2 / 4 / M
ELEC -0034	BEng (Hons) in Sustainable Energy Engineering (WD_ESERG_B)	2 / 4 / M
	BEng (Hons) (WD_ETRIC_B)	2 / 4 / M
ELEC -0034	BEng in Electrical Engineering (WD_EELCT_D)	3 / 6 / M

Indicative Content

- Cables & cable sizing, switchboard & switchgear design
- Electrical Containment Systems
- Fire Alarm Systems
- Lightning protection & UPS systems
- Testing & Certification

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the electrical systems that are found in modern buildings & plant.
2. Apply standard calculation techniques necessary for the design of electrical systems within a modern building.
3. Recognise and understand the comparative cost and operational benefits of different types of electrical services and systems within modern buildings.
4. Explain the procedures involved in designing the electrical services for a building/ plant.
5. Apply electrical services design techniques.

Learning and Teaching Methods

- Lectures, presentations, reading and problem solving in class.
- Site visits and (on site) presentation of specialist equipment and topics
- Practical design Exercises
- Case study based on the design and layout of electrical services and systems

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2
Continuous Assessment	40%	
Assignment	40%	3,4,5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Electro Tech Council of Ireland, I. *National Rules for Electrical Installations*. 4th ed.. UK: ., 2008.
- Porges, F. and B. Rigby. *Design of Electrical Services for Buildings*. 4th ed.. UK: ., 2005.

Supplementary Material

- Cook, P. *Electrical Installations Designs*. 4th ed.. UK: Wiley, 2013.
- Henshaw, M. *Protection against Lightning*. UK: BSI, 2007.
- Kitcher, C. *Practical Guide to Inspection Testing 2013*. 4th ed.. UK: Routledge, 2015.
- Koffler, R. *The Power Protection Guide*. UK: Stillwater Publishing, 2006.

Requested Resources

- ENGINEERING LAB: Electrical